Deep Learning-Based Tourist Spot Recognition and Categorization For Tourism Enhancement

Submitted By

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Introduction:

- Tourism Vitality: A Key Economic Driver in Bangladesh, Fueling GDP Growth, Job Creation, and Revenue
- Tourism enhances the global visibility of Bangladesh and tourism sector provides employment opportunities.
- We need clear guidelines and smart solutions to introduce tourist spots and improve the tourism sector. Our research is focused on addressing this need

Hypothesis :

- Automatically recognition and categorization of tourist spots
- Address the issue of limited information about travel destinations on social media
- Utilization of deep learning models
- Significantly improve the accuracy and accessibility of tourist spot recognition and categorization

Objectives :

- Dataset development and contribution
- Develop a deep learning model to identify tourist spots from images.
- Creating tourist spot classification system
- Tourist spot image feature extraction implementation
- Analyzing real-world robustness of deep learning models in changing conditions
- Promoting travel for positive impact in tourism

Literature Review

Paper 01	**“Tourist Spot Recognition using Machine Learning Algorithms”** by Pranta Roy, Jahanggir Hossain Setu, Afrin Nahar Binti, na Yeasmin Koly and Nusrat JahanFarja	➢ using CNN, SVM, LSTM, KNN, RNN to identify tourist attractions from images ➢ utilizes the processing power of modern devices to function effectively without requiring an internet connection. ➢ Highest accuracy (CNN) – 97%
Paper 02	**“Deep learning based real-time tourist spots detection and recognition mechanism”** by Yen-Chiu Chen, Kun-Ming Yu, Tzu-Hsiang Kao and Hao-Lun Hsieh	➢ Quickly recognize and identify tourist spots from pictures taken or images found online. ➢ Models used : YOLOV3, Faster R-CNN, SSD ➢ Higher accuracy(88.63%) & Faster detection (4.5s) - YOLOV3

Literature Review(Cont.)

Paper 03	**"Tourist spot classification using convolution neural network"** by Tejaswini Bhosale, S.Pushkar	➢ Utilizing CNN for correctly classifying tourist spots ➢ Trained on five broad categories, including heritage, beaches, pilgrimage sites, nature & wildlife, Ayurveda & spiritual spots. ➢ Superior Performance: CNN model (91.4% accuracy) compared to 24.9% for HOG and 59.1% for SIFT.
Paper 04	**"Deep learning and Internet of Things for tourist attraction recommendations in smart cities"** by Juan Carlos Cepeda-Pacheco, Mari Carmen Domingo	➢ Combining IoT technology with deep learning techniques to create a recommendation system for tourist attractions. ➢ Multilayer Perceptron (MLP)

Methodology :

01 Data collection and then pre-processing the data through multiple ways

02 Splitting the processed data into train and test set

03 Using different deep learning models to train the dataset and subsequently testing them

04 Identifying the tourist spots and classifying them

05 Evaluate the accuracy, precision, recall, and performance of each deep learning model

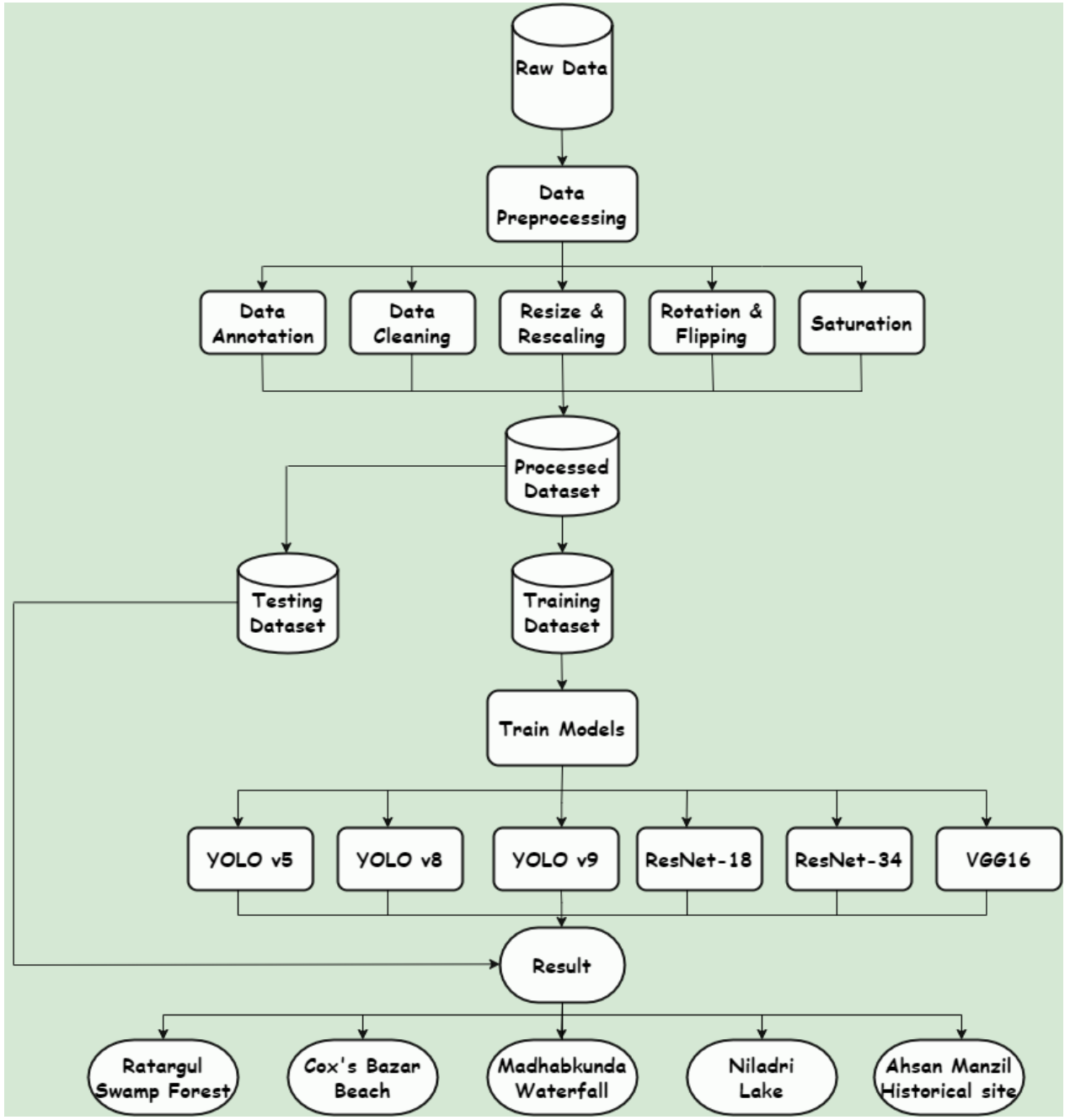


Fig: Workflow Diagram

Dataset

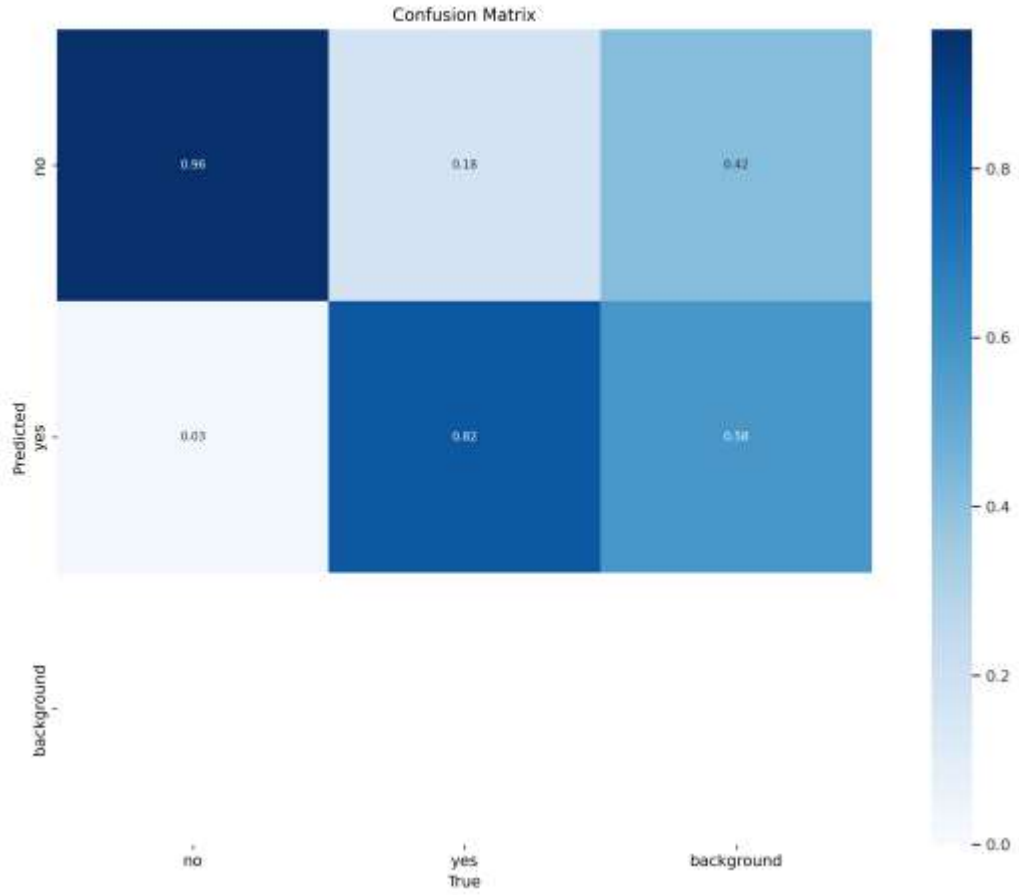
For Detection :

Classes	Number Of Images
Tourist Spot (Yes)	2030
Non-Tourist Spot (No)	1852

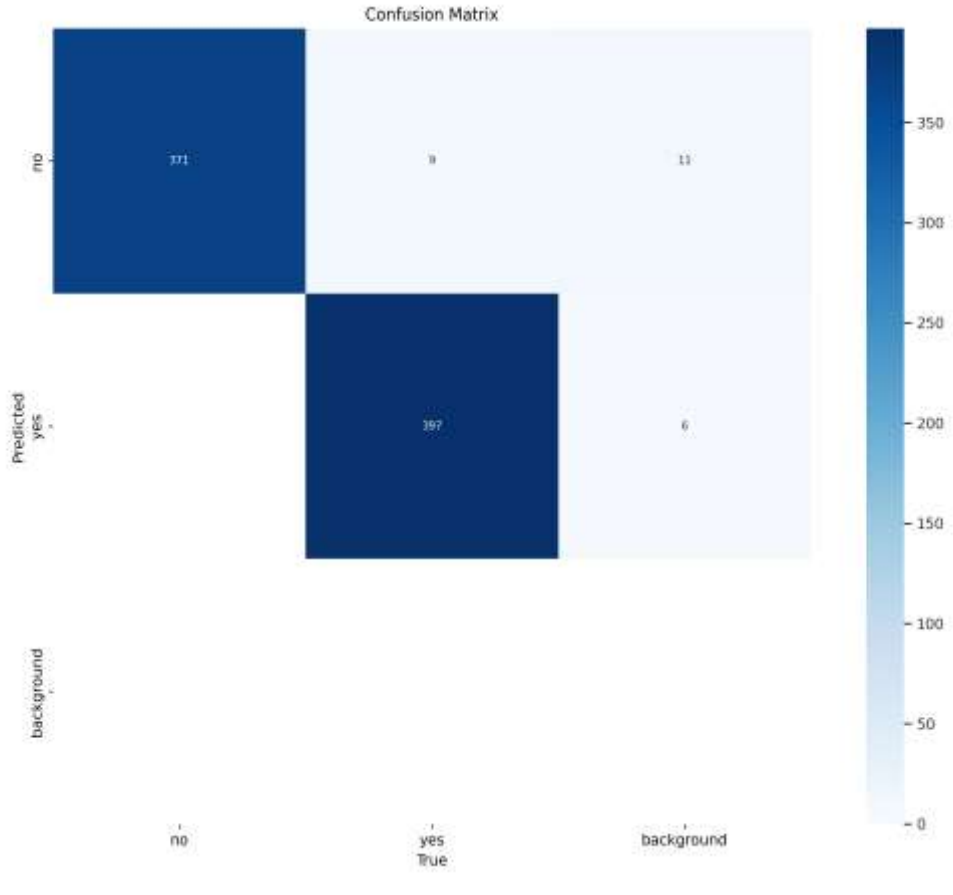
For Classificatio

Classes	Number Of Images
Dibir Haor	209
Hum Hum Waterfall	194
Lalakhal	197
Madhabkunda	177
Niladri	193
Ratargul	199
Shada Pathor	210
Shimul Bagan	197
Tanguar Haor	207
Cox's Bazar	209
Saint Martin Island	212
Chittagong War Cemetery	193
Kaptai Lake	202
Guliakali Sea Beach	205
Patenga Sea Beach	207
Mahasthangarh	210
Chimbuk Hill	165
Nafakhum Falls	173
Kuakata Sea Beach	217
Ahsan Manzil Museum	265
Jaflong	190

Result & Discussion:

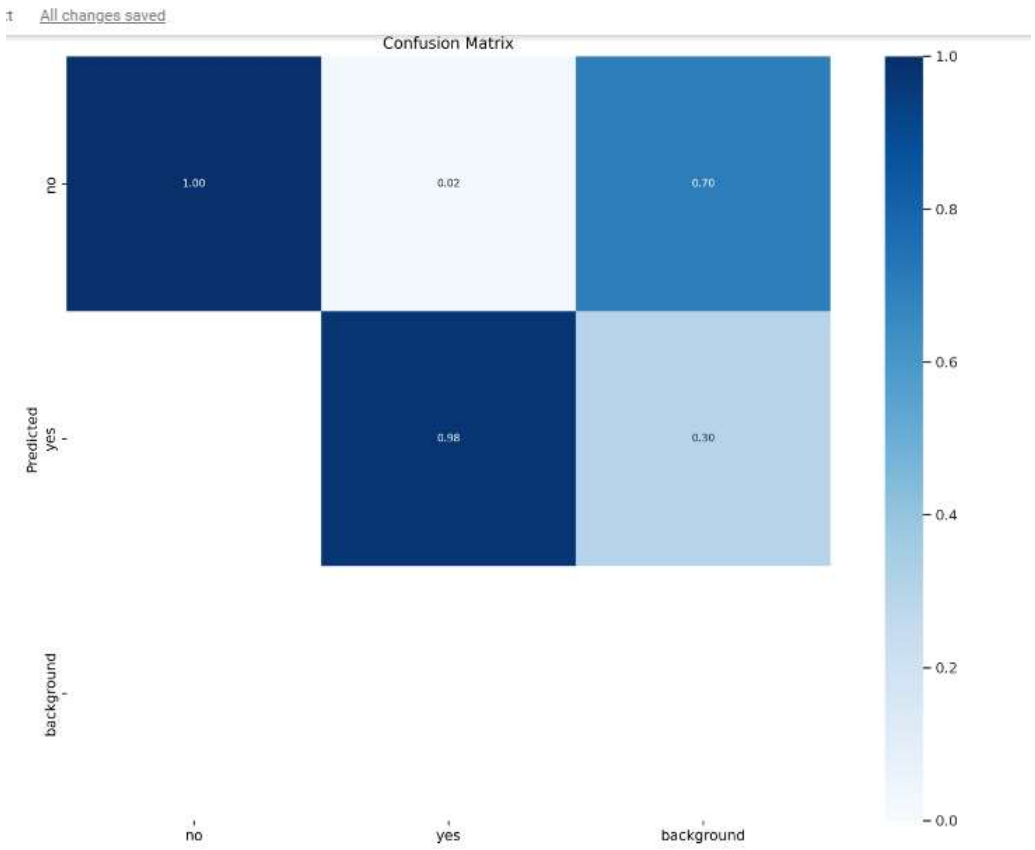


CM For YOLOV5

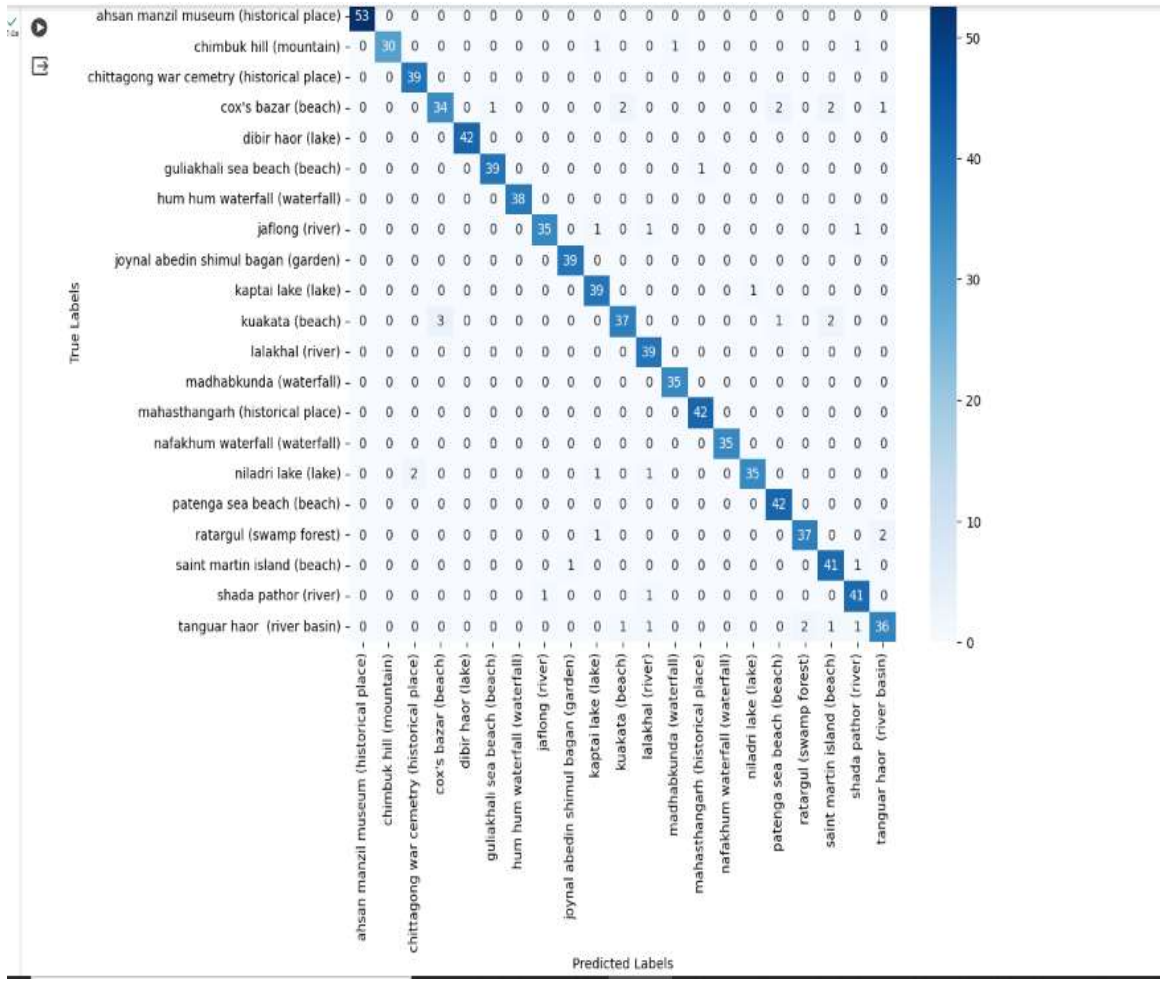


CM For YOLOV8

Result & Discussion:

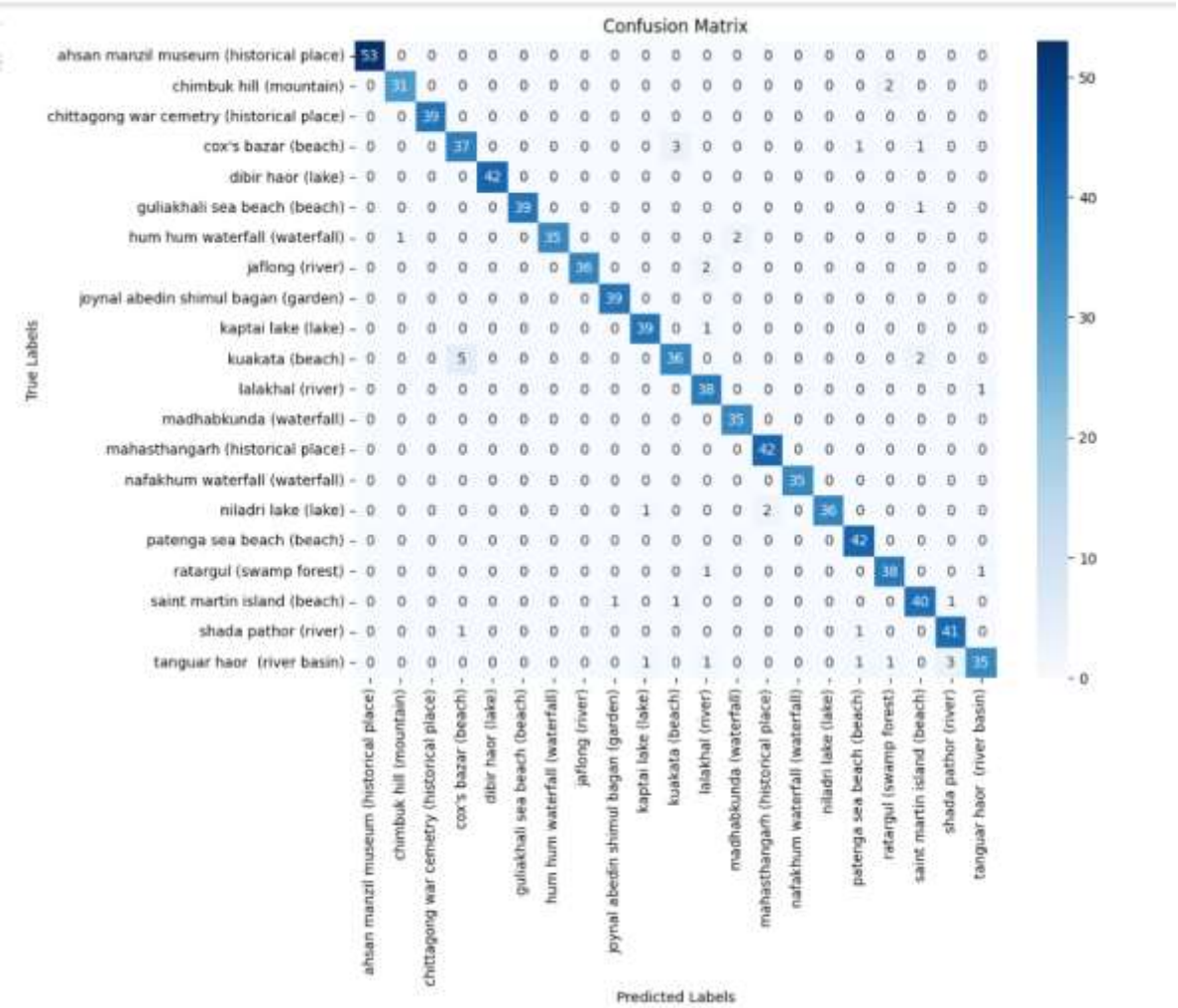


CM For YOLOV9

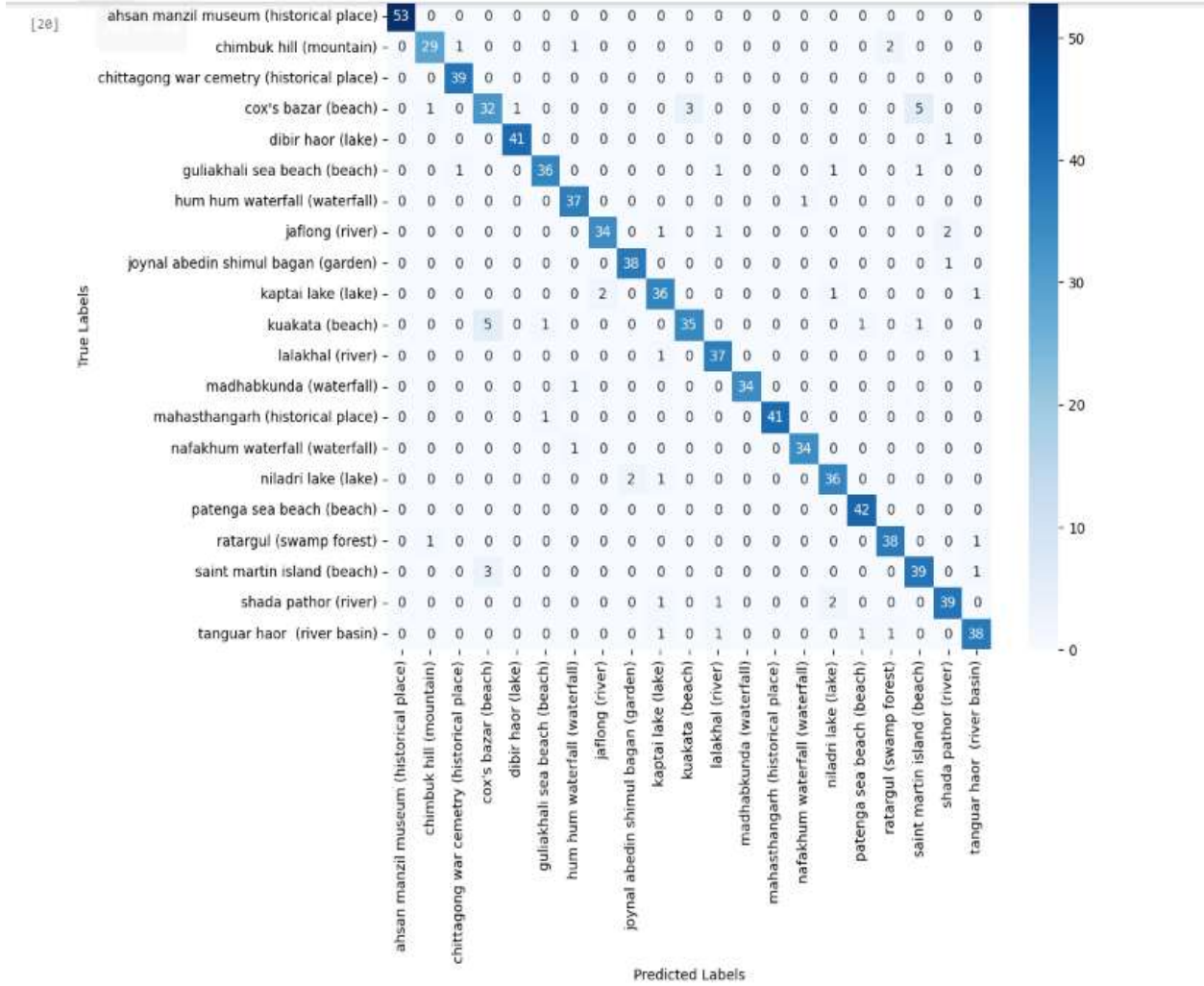


CM For ResNet-18

Result & Discussion:



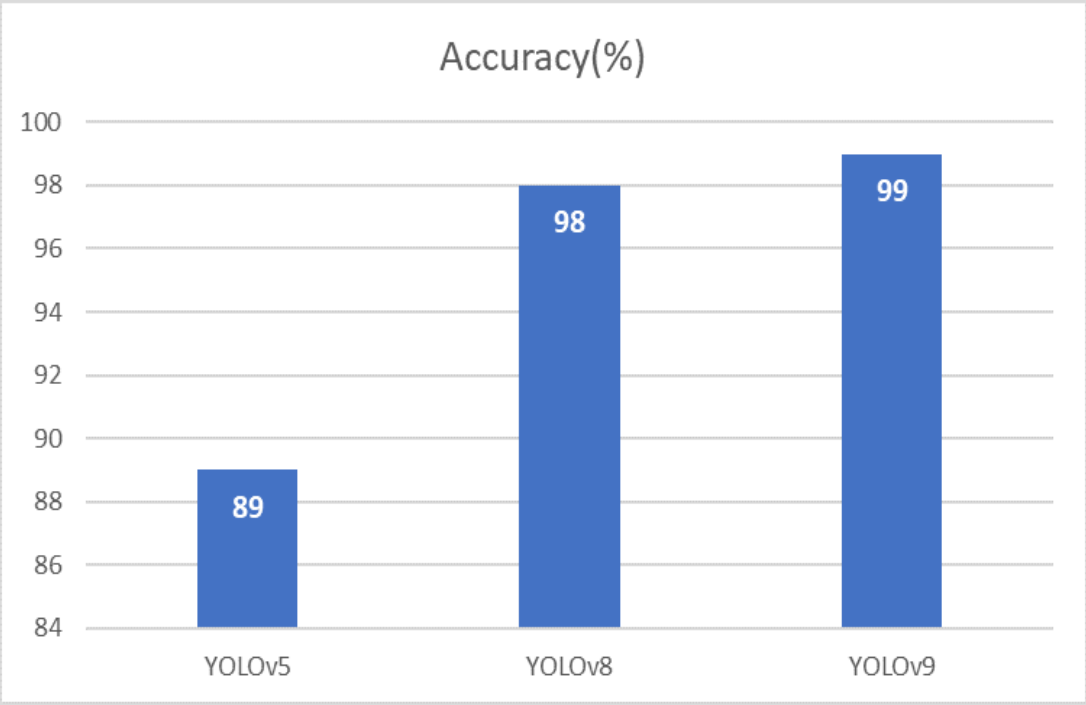
CM For ResNet-34



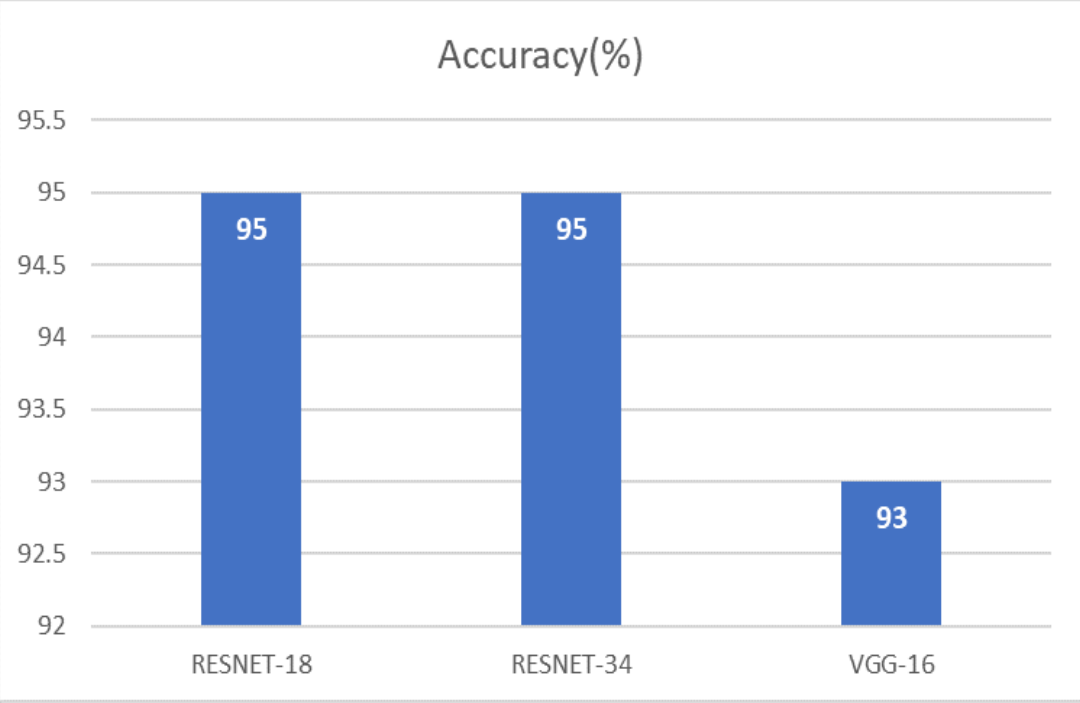
CM For VGG-16

Result & Discussion:

Accuracy for tourist spot detection :

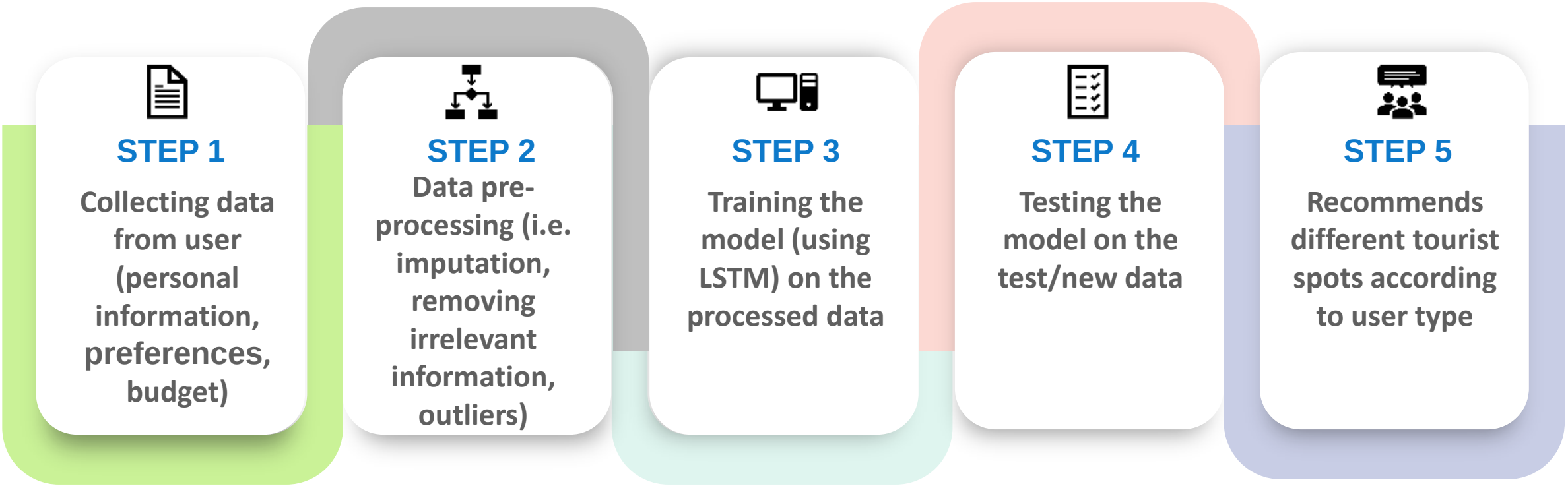


Accuracy for tourist spot classification :



Future Work:

Building a recommendation system



Conclusion:

- The above discussion states that various tourist spots in Bangladesh are categorized and labeled based on provided images using deep learning models.

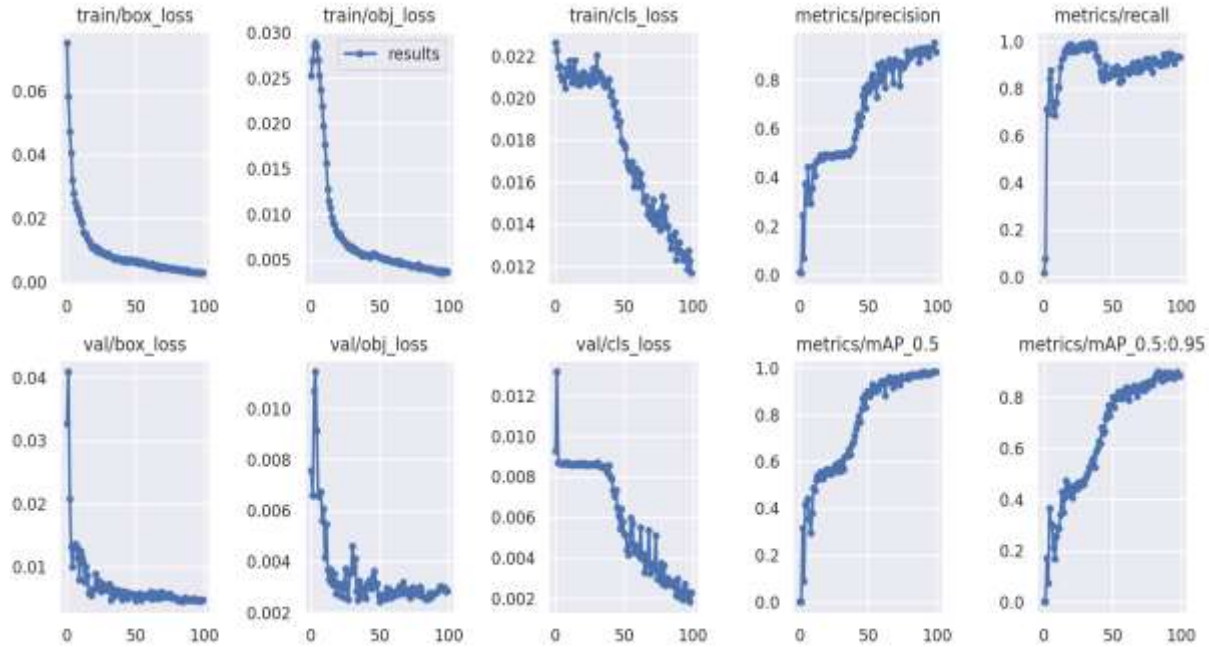
- We prioritize training deep learning models by exposing them to diverse images, with the goal of improving their accuracy in identifying and categorizing tourist spots, surpassing previous methods.

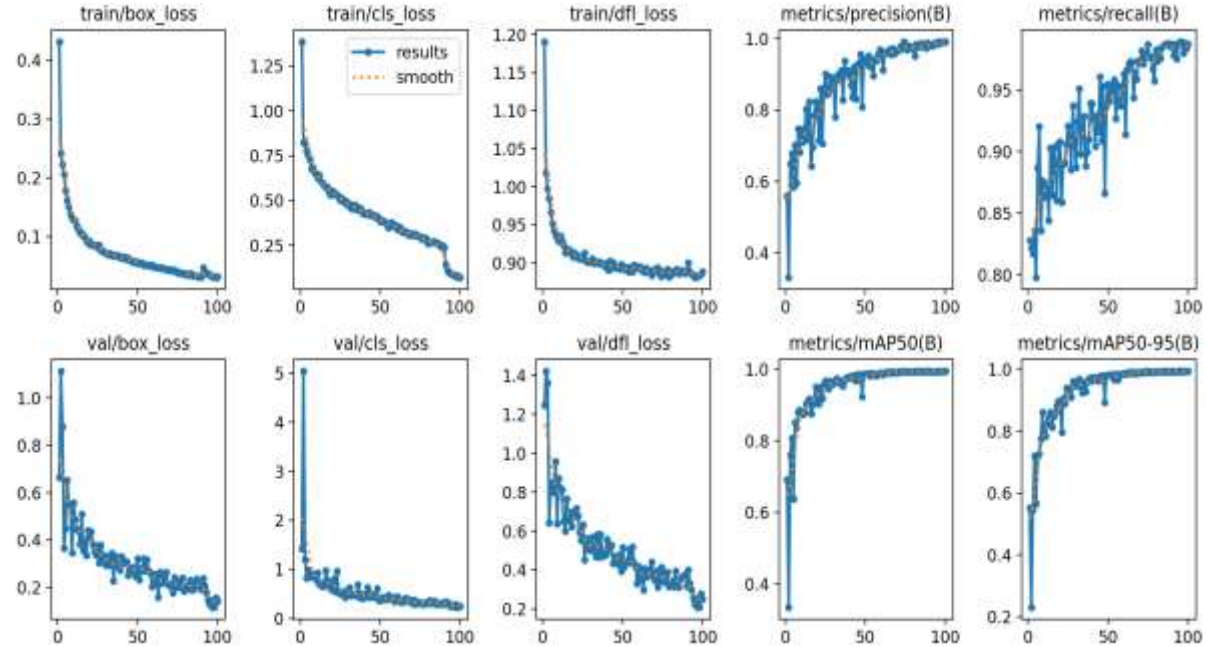
- Furthermore, our future efforts will involve developing recommendations for tourist spots based on individual interests.

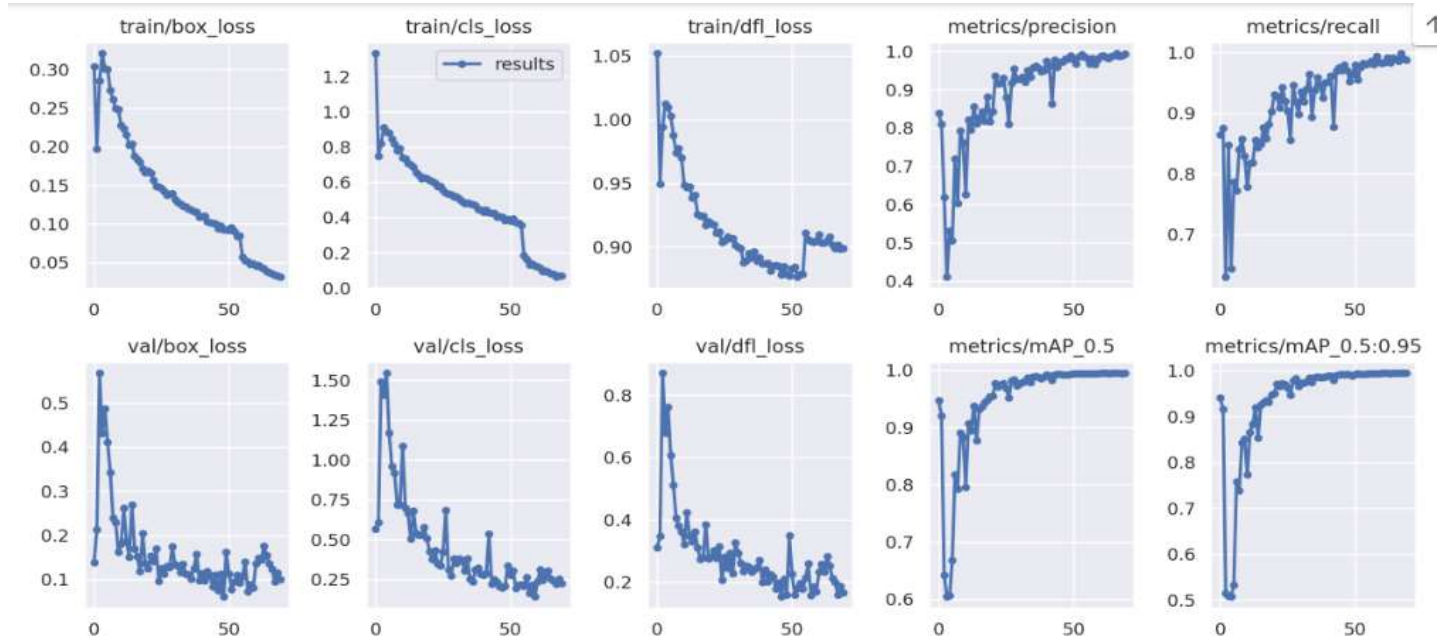
Thank You



Questions


train/box_loss
train/obj_loss
results
train/cls_loss
metrics/precision
metrics/recall
val/box_loss
val/obj_loss
val/cls_loss
metrics/mAP_0.5
metrics/mAP_0.5:0.95


train/box_loss
train/cls_loss
results
smooth
train/dfl_loss
metrics/precision(B)
metrics/recall(B)
val/box_loss
val/cls_loss
val/dfl_loss
metrics/mAP50(B)
metrics/mAP50-95(B)


train/box_loss
train/cls_loss
results
train/dfl_loss
metrics/precision
metrics/recall
val/box_loss
val/cls_loss
val/dfl_loss
metrics/mAP_0.5
metrics/mAP_0.5:0.95